Closed-loop cell therapy

Technical documentation of the process model and of the work done so far (days 1 to 3)

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Repository `closed-loop-cell-therapy`, commit `5e6611b`, version 0.1.0

In one paragraph

The repository simulates one autologous CAR-T batch that grows for about ten days in a perfused rocking bag. The heart of it is `process/model.py`: seven ordinary differential equations for two T-cell populations (memory-like and effector-like), dead cells, glucose, lactate, bicarbonate and dissolved oxygen. Around the model sit an inner control loop (the rocker's own pH and DO controller), a sensor layer that turns the true state into noisy, drifting, delayed measurements, and a runner that advances everything one five-minute control interval at a time. The question behind the project: the memory-cell share falls as the culture gets longer, but it is normally measured only at harvest. Can it be estimated during the run, with uncertainty, and used to decide when to sample and when to harvest?

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1 What exists so far: the three steps

The README plans the project as twelve day-sized steps. The first three are done. The second git commit covers days 1 and 2 together; the third commit is day 3.

Day	Step	What it added	Main files
1	Repo, architecture, two-population model	Package layout, ISA-95 architecture note, the seven-state ODE model with its rate laws and observables, parameters with units, status and sources.	`process/model.py`, `process/parameters.py`, `docs/architecture.md`, `docs/calibration.md`
2	Patients A and B, clock, rocking-bag controller	Two calibrated patients (a fit donor and a slow donor), the PI controller, the rocker's pH, DO and perfusion loops with hard limits, the batch runner, the accelerated clock, two demo scripts and tests.	`control/pid.py`, `control/rocker.py`, `sim/runner.py`, `sim/clock.py`
3	Sensor layer	Online pH and DO patches and load cell (noise, drift, update rate, faults); the controller now acts on readings, not truth; at-line cell counter and metabolite analyser with turnaround; closed sampling that costs cells; flow cytometry at harvest; soft sensors for oxygen uptake and acid production.	`sensors/online.py`, `sensors/atline.py`, `sensors/soft.py`, `docs/sensors.md`

All 18 tests pass on this commit. The figures in this document were produced by running the current code.

2 Repository map

```
closed-loop-cell-therapy/
  pyproject.toml        package 'cltherapy'; numpy, scipy, matplotlib; extras for later steps
  src/cltherapy/
    process/            the simulated world (ground truth); knows nothing about sensors or control
      parameters.py     every parameter with unit, status and source; patients A and B
      model.py          states, inputs, rate laws, ODE right-hand side, observables
    control/            the rocker's embedded firmware
      pid.py            PI controller with output limits and anti-windup
      rocker.py         pH by CO2; DO by rock rate then O2; perfusion schedule; hard limits
    sensors/            what the platform can actually see
      online.py         optical pH and DO patches, load cell: noise, drift, hold, faults
      atline.py         cell counter, metabolite analyser, flow cytometry at harvest
      soft.py           oxygen uptake rate and acid production from online signals
    sim/
      runner.py         Batch: control step, ODE integration, history, sampling, CSV
      clock.py          simulated hours per wall-clock second, for live mode later
  scripts/              run_two_patients.py, step_response.py, sensor_demo.py -> outputs/
  tests/                test_process, test_control, test_sensors, test_clock
  docs/                 architecture.md, calibration.md, sensors.md
```

Dependencies point one way. *process* depends on nothing; *control* imports only the `Inputs` dataclass from the model; *sensors* reuse the model's k_La correlation; *sim* ties them together:

```
parameters --> model (truth) --> sensors (noisy view) --> rocker controller
                  ^                                            |
                  +---------- Inputs (CO2, O2, rpm, VVD) <-----+
```

3 The process model (`model.py`)

3.1 Conventions and units

Time t is in hours. Cell densities are in 10^6 cells/mL. Concentrations are in mM. Cell-specific rates q are in pmol/cell/h. These units make a specific rate times a density directly a volumetric rate in mM/h:

$$1\,\text{pmol/cell/h} \times 10^6\,\text{cells/mL} = \frac{10^{-12}\,\text{mol} \times 10^6}{10^{-3}\,\text{L}\cdot\text{h}} = 10^{-3}\,\text{mol/L/h} = 1\,\text{mM/h}. \tag{1}$$

The bag volume is constant at 1 L: perfusion removes exactly what it adds, and a cell-retention filter keeps all cells, live and dead, in the bag. Temperature is held at 37 °C.

3.2 State vector

$$x = \begin{pmatrix} M & E & X_d & G & L & B & O \end{pmatrix}^\top \tag{2}$$

Index	Symbol	Meaning	Unit	Initial value $x(0)$
0	M	memory-like viable cells	10^6 cells/mL	$X_0\, m_0$
1	E	effector-like viable cells	10^6 cells/mL	$X_0\,(1 - m_0)$
2	X_d	dead cells (retained, never lysed)	10^6 cells/mL	$X_0\,(1 - v_0)/v_0$
3	G	glucose	mM	$G_{\text{in}} = 20$
4	L	lactate	mM	$L_{\text{in}} = 0$
5	B	bicarbonate	mM	$B_{\text{in}} = 24$
6	O	dissolved oxygen	mM	$C^*_{\text{air}}\, y_{O_2}/0.21$

Here X_0 is the seed density, m_0 the seed memory fraction and v_0 the seed viability, so that the initial viability is $(M + E)/(M + E + X_d) = v_0$. The order is fixed by `STATE_NAMES`. Every function first clips the state to $x \geq 0$, so a solver overshoot below zero never produces a negative rate.

3.3 Inputs

The inputs u are set by the controller once per control interval and held constant while the ODEs are integrated (a zero-order hold).

Field	Symbol	Meaning	Default	Set by
`perfusion_vvd`	VVD	fresh medium, vessel volumes per day	0	perfusion schedule
`rock_rpm`	n	rocking rate	15 rpm	DO loop (first half)
`rock_angle_deg`	θ	rocking angle	6°	fixed
`y_O2`	y_{O_2}	O_2 fraction in headspace gas	0.21	DO loop (second half)
`y_CO2`	y_{CO_2}	CO_2 fraction in headspace gas	0.05	pH loop

The dilution rate used in the balances is

$$D = \frac{VVD}{24} \quad [1/\text{h}]. \tag{3}$$

3.4 Physical sub-models

3.4.1 Oxygen transfer: `kla()`

Rocking creates a wave and the gas–liquid surface that oxygen crosses. The volumetric transfer coefficient is a power law normalised to a reference setting:

$$k_L a(n,\theta) = k_L a_{\mathrm{ref}} \left(\frac{n}{n_{\mathrm{ref}}}\right)^{\alpha_n} \left(\frac{\theta}{\theta_{\mathrm{ref}}}\right)^{\alpha_\theta}, \tag{4}$$

with $k_L a_{\mathrm{ref}} = 2.0/\mathrm{h}$, $n_{\mathrm{ref}} = 15\,\mathrm{rpm}$, $\theta_{\mathrm{ref}} = 6°$, $\alpha_n = 0.8$ and $\alpha_\theta = 0.5$. At $\theta = 6°$ this gives 0.96/h at the minimum 6 rpm, 2.0/h at 15 rpm and 3.0/h at the maximum 25 rpm. The oxygen concentration in equilibrium with the gas is

$$C^* = C^*_{\mathrm{air}} \frac{y_{\mathrm{O_2}}}{0.21}, \qquad C^*_{\mathrm{air}} = 0.19\,\mathrm{mM}, \tag{5}$$

and dissolved oxygen is reported as $\mathrm{DO} = 100\,O/C^*_{\mathrm{air}}$ in % air saturation.

3.4.2 pH: `ph_from()`

pH follows from the bicarbonate buffer and the headspace CO_2 through Henderson–Hasselbalch at 37 °C:

$$\mathrm{pH} = \mathrm{p}K_a + \log_{10} \frac{B}{s\,p_{\mathrm{CO_2}}}, \qquad p_{\mathrm{CO_2}} = \kappa\, y_{\mathrm{CO_2}}, \tag{6}$$

with $\mathrm{p}K_a = 6.1$, $s = 0.0307\,\mathrm{mM/mmHg}$ and $\kappa = 713\,\mathrm{mmHg}$ per unit gas fraction. Fresh medium ($B = 24\,\mathrm{mM}$) under 5% CO_2 gives

$$\mathrm{pH} = 6.1 + \log_{10} \frac{24}{0.0307 \times 713 \times 0.05} = 6.1 + \log_{10} 21.9 = 7.44.$$

More CO_2 lowers pH; lactic acid consumes bicarbonate and also lowers it. pH is not a state: it is recomputed from B and $y_{\mathrm{CO_2}}$ wherever it is needed. Floors $y_{\mathrm{CO_2}} \geq 10^{-4}$ and $B \geq 10^{-3}\,\mathrm{mM}$ keep the logarithm finite.

3.5 Rate laws: `rates()`

`rates()` computes every rate at one instant. It is separate from the right-hand side because the soft sensors and the observables reuse the same quantities.

3.5.1 Growth factors

Growth is a maximum rate multiplied by five dimensionless factors in $[0, 1]$:

$$f = f_{\mathrm{lag}}(t)\; f_G(G)\; f_L(L)\; f_{\mathrm{pH}}(\mathrm{pH})\; f_{\mathrm{O_2}}(O), \tag{7}$$

$$f_{\mathrm{lag}}(t) = 1 - e^{-t/\tau_{\mathrm{lag}}}, \qquad \tau_{\mathrm{lag}} = 24\,\mathrm{h}, \tag{8}$$

$$f_G(G) = \frac{G}{K_G + G}, \qquad K_G = 0.5\,\mathrm{mM}, \tag{9}$$

$$f_L(L) = \frac{K_{iL}}{K_{iL} + L}, \qquad K_{iL} = 40\,\mathrm{mM}, \tag{10}$$

$$f_{\mathrm{pH}}(\mathrm{pH}) = \begin{cases} 1, & \mathrm{pH} \geq \mathrm{pH}_{\mathrm{low}}, \\ \max\!\big(0,\; 1 - 4\,(\mathrm{pH}_{\mathrm{low}} - \mathrm{pH})\big), & \mathrm{pH} < \mathrm{pH}_{\mathrm{low}}, \end{cases} \qquad \mathrm{pH}_{\mathrm{low}} = 6.95, \tag{11}$$

$$f_{\mathrm{O_2}}(O) = \frac{O}{K_{\mathrm{O_2}} + O}, \qquad K_{\mathrm{O_2}} = 0.005\,\mathrm{mM}. \tag{12}$$

In words: cells need about a day after activation before dividing at full speed; glucose limits growth only when nearly exhausted; lactate halves growth at 40 mM; pH has no effect above 6.95 and stops growth linearly by pH 6.70; oxygen limits growth only below a few percent air saturation (K_{O_2} is about 2.6% of air saturation and is hard-coded, see section 10).

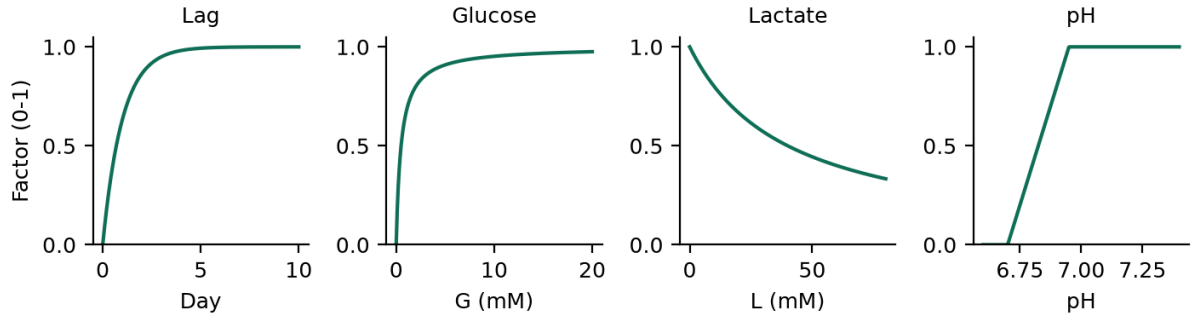


Figure 1: The growth factors (8)–(11). f_{O_2} is not shown; it is close to 1 above about 10% air saturation.

3.5.2 Growth

$$\mu_M = \mu_{\max,M}\, f, \qquad \mu_E = \mu_{\max,E}\, f. \tag{13}$$

Effector-like cells grow faster than memory-like cells: for patient A, $\mu_{\max,E} = 0.0306/\text{h}$ and $\mu_{\max,M} = 0.0205/\text{h}$, doubling times $\ln 2/\mu_{\max}$ of about 23 h and 34 h at $f = 1$. Both populations share the same environmental factor f.

3.5.3 Differentiation from memory to effector

$$k_{\text{diff}}(t) = k_{\text{diff},0}\, 2^{\,t/T_d}\, \big(1 - r_a\, a\big), \tag{14}$$

with $k_{\text{diff},0} = 0.0018/\text{h}$, $T_d = 210$ h (the rate doubles about every 8.75 days), $r_a = 0.5$, and $a \in [0, 1]$ the patient's activation strength. The rising rate encodes the observation that longer culture pushes cells towards a differentiated phenotype. Strong activation slows differentiation: at $a = 1$ the rate is halved.

3.5.4 Metabolism

$$q_G = \big(q_{G,M}\, M + q_{G,E}\, E\big)\, f_G(G), \tag{15}$$

$$r_L = Y_{L/G}\, q_G\, \beta(t), \qquad \beta(t) = 1 + b\, a\, e^{-t/\tau_b}, \tag{16}$$

$$\text{OUR} = \big(q_{O_2,M}\, M + q_{O_2,E}\, E\big)\, f_{O_2}(O). \tag{17}$$

q_G is the volumetric glucose uptake, r_L the lactate production and OUR the oxygen uptake rate, all in mM/h. Effector-like cells are more glycolytic ($q_{G,E} = 0.060$ versus $q_{G,M} = 0.035$ pmol/cell/h); memory-like cells rely more on oxidative phosphorylation ($q_{O_2,M} = 0.030$ versus $q_{O_2,E} = 0.020$ pmol/cell/h). Lactate is made at $Y_{L/G} = 1.6$ mol per mol glucose. $\beta(t)$ is the early glycolytic burst: activation adds up to $b = 1$ (double lactate) at $t = 0$, decaying with $\tau_b = 36$ h. Glucose uptake is limited only by glucose itself, not by f_{lag}, f_{pH} or f_L.

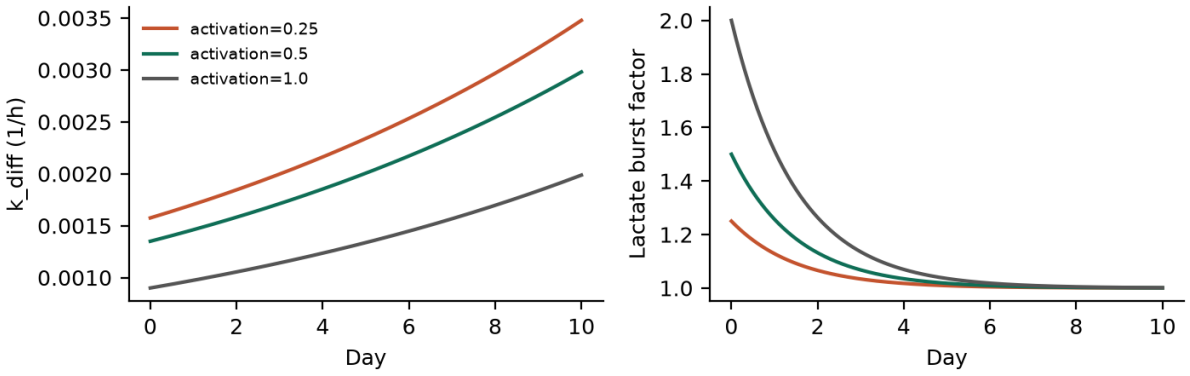


Figure 2: Left: differentiation rate (14) for three activation strengths (patient B uses $a = 0.25$, patient A $a = 0.5$). Right: the lactate burst factor $\beta(t)$ of (16).

The modelling hypothesis

The single patient parameter a does two things: it raises early lactate production through $\beta(t)$ and it slows later differentiation through (14). Together they encode the correlation reported by Jagannathan et al. (SMART, *Cell Reports Methods* 2026) that early lactate goes with more central memory at harvest. This gives the future EnKF a measurable early signal (lactate, acid production) that carries information about a hidden late quantity (the memory share). The code calls this structure a hypothesis to test.

3.6 The ODE system: `rhs()`

With death rates $\delta_M = 0.001/\text{h}$ and $\delta_E = 0.003/\text{h}$:

$$\frac{dM}{dt} = \left(\mu_M - k_{\text{diff}} - \delta_M\right) M, \tag{18a}$$

$$\frac{dE}{dt} = \left(\mu_E - \delta_E\right) E + k_{\text{diff}}\, M, \tag{18b}$$

$$\frac{dX_d}{dt} = \delta_M M + \delta_E E, \tag{18c}$$

$$\frac{dG}{dt} = D\,(G_{\text{in}} - G) - q_G, \tag{18d}$$

$$\frac{dL}{dt} = D\,(L_{\text{in}} - L) + r_L, \tag{18e}$$

$$\frac{dB}{dt} = D\,(B_{\text{in}} - B) - \gamma\, r_L, \tag{18f}$$

$$\frac{dO}{dt} = k_L a\left(C^* - O\right) - \text{OUR}. \tag{18g}$$

Read line by line:

- **Cells** (18a)–(18b) have no dilution term: the perfusion filter retains them. Memory cells leave their pool by differentiation and death; effector cells gain what memory cells lose by differentiation.
- **Dead cells** (18c) only accumulate, so viability drifts down slowly.
- **Glucose, lactate and bicarbonate** (18d)–(18f) are perfused-reactor balances: inflow at fresh-medium concentration, outflow at bag concentration, plus consumption or production.
- **Bicarbonate** absorbs only a share $\gamma = 0.4$ of the lactic acid; the rest is taken up by HEPES and other buffers, lumped into this one number.
- **Oxygen** (18g) enters only through the surface and is not carried by perfusion.

3.7 Why the memory share falls

Let $X = M + E$ be the viable density and $m = M/X$ the memory share. Then

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{1}{X}\frac{\mathrm{d}M}{\mathrm{d}t} - \frac{M}{X^2}\frac{\mathrm{d}X}{\mathrm{d}t} = m\left[\frac{1}{M}\frac{\mathrm{d}M}{\mathrm{d}t} - \frac{1}{X}\frac{\mathrm{d}X}{\mathrm{d}t}\right].$$

From (18a)–(18b),

$$\frac{1}{M}\frac{\mathrm{d}M}{\mathrm{d}t} = \mu_M - k_{\mathrm{diff}} - \delta_M, \qquad \frac{1}{X}\frac{\mathrm{d}X}{\mathrm{d}t} = m\,(\mu_M - \delta_M) + (1-m)\,(\mu_E - \delta_E),$$

where the k_{diff} terms cancel in X because differentiation only moves cells between pools. Subtracting,

$$\boxed{\frac{\mathrm{d}m}{\mathrm{d}t} = \underbrace{m(1-m)\Big[(\mu_M - \delta_M) - (\mu_E - \delta_E)\Big]}_{\text{selection: effectors out-grow memory}} - \underbrace{k_{\mathrm{diff}}(t)\,m}_{\text{conversion}}} \tag{19}$$

Two mechanisms lower the share. *Selection*: with (13) the bracket equals $-(\mu_{\max,E} - \mu_{\max,M})\,f + (\delta_E - \delta_M)$, negative whenever the culture is growing, so effectors out-grow memory cells. *Conversion*: memory cells turn into effectors at a rate that rises with time. The selection term scales with f, so anything that slows growth (lag, a slow donor, lactate) also slows the loss of memory; the conversion term does not. Section 10 shows why this matters for patient B.

3.8 Observables: `observables()`

Key	Definition	Who can see it
`viable_density`	$X = M + E$	at-line cell counter (sparse)
`viability`	$X/(X + X_d)$	at-line cell counter
`memory_fraction`	$m = M/X$	**hidden**: flow cytometry at harvest only
`total_viable_cells`	$X \cdot V \cdot 10^3$ in 10^6 cells (the dose)	derived
`glucose`, `lactate`	G, L	at-line metabolite analyser
`pH`	equation (6)	optical patch, online
`DO_pct`	$100\,O/C^*_{\mathrm{air}}$	optical patch, online
`OUR`	equation (17)	soft sensor
`acid_production`	r_L, equation (16)	soft sensor

4 Parameters and patients (`parameters.py`)

Parameters are split into two frozen dataclasses. `PatientParams` holds what differs between patients; a future EnKF will learn these by augmenting them into its state. `ProcessParams` holds the cells' shared biology, the medium and the bag. `PARAM_INFO` attaches a unit, a status and a source to each calibration-relevant parameter. The status is one of **anchor** (a published number the model must reproduce), **placeholder** (plausible, to be replaced by a fit to digitised literature data) or **assumed** (a modelling choice).

4.1 Patient parameters

Code name	Symbol	Patient A	Patient B	Unit	Status / source
`mu_max_M`	$\mu_{\max,M}$	0.0205	0.0172	1/h	placeholder; Costariol 2019, Egan 2025
`mu_max_E`	$\mu_{\max,E}$	0.0306	0.0256	1/h	placeholder; same
`k_diff0`	$k_{\text{diff},0}$	0.0018	0.0018	1/h	placeholder; fitted to memory anchors
`activation`	a	0.5	0.25	–	assumed; SMART 2026
`seed_density`	X_0	0.5	0.5	10^6 cells/mL	–
`seed_memory_fraction`	m_0	0.45	0.45	–	placeholder; Cadinanos-Garai 2025
`seed_viability`	v_0	0.95	0.85	–	–

Patient A is the fit donor. Patient B is a slow donor: growth rates are scaled so its day-9 density is 40% below patient A's, within the 30–50% gap SMART (2026) reported for patient cells at day 12.

4.2 Process parameters

Group	Code name	Symbol	Value	Status / note
growth	`lag_tau_h`	τ_{lag}	24 h	activation lag
	`diff_time_scale_h`	T_d	210 h	placeholder
	`activation_diff_reduction`	r_a	0.5	
	`k_death_M`, `k_death_E`	$\delta_M,\ \delta_E$	0.001, 0.003/h	
	`K_glc`	K_G	0.5 mM	Monod constant
	`K_i_lac`	K_{iL}	40 mM	lactate inhibition
	`pH_opt_low`	pH_{low}	6.95	
	(hard-coded)	K_{O_2}	0.005 mM	in `rates()`
metabolism	`q_glc_M`, `q_glc_E`	$q_{G,M},\ q_{G,E}$	0.035, 0.060	pmol/cell/h; placeholder
	`Y_lac_glc`	$Y_{L/G}$	1.6	mol/mol; Costariol 2019
	`early_lac_boost`	b	1.0	
	`early_lac_tau_h`	τ_b	36 h	
	`q_O2_M`, `q_O2_E`	$q_{\text{O}_2,M},\ q_{\text{O}_2,E}$	0.030, 0.020	pmol/cell/h; SMART 2026
medium	`glc_in`, `lac_in`	$G_{\text{in}},\ L_{\text{in}}$	20, 0 mM	placeholder
	`hco3_in`	B_{in}	24 mM	assumed
	`acid_to_bicarb`	γ	0.4	assumed; Egan 2025 pH traces
bag	`volume_L`	V	1.0 L	
	`O2_sat_air`	C^*_{air}	0.19 mM	at 37 °C
	`kLa_ref`	k_La_{ref}	2.0/h	placeholder; measure per bag
	`rock_ref_rpm`, `angle_ref_deg`	$n_{\text{ref}},\ \theta_{\text{ref}}$	15 rpm, 6°	
	`kLa_rpm_exp`, `kLa_angle_exp`	$\alpha_n,\ \alpha_\theta$	0.8, 0.5	
	`pCO2_per_frac`	κ	713 mmHg	
	`pKa_bicarb`	$\text{p}K_a$	6.1	
	`CO2_solubility`	s	0.0307 mM/mmHg	

4.3 Anchors

`MEMORY_ANCHORS` = $\{5 : 0.25,\ 10 : 0.115\}$ is the stem-cell memory share of CAR$^+$ cells at days 5 and 10 across six healthy donors (Cadinanos-Garai et al., *Molecular Therapy* 2025). $k_{\text{diff},0}$, T_d, $\mu_{\max,M}$ and $\mu_{\max,E}$ were fitted so that patient A meets both anchors and reaches about 20×10^6 cells/mL on day 9. The anchors are therefore met by construction: they are a calibration check, not a validation.

5 What the model produces

A ten-day closed-loop run of each patient with the default controller and perfusion schedule, with no sensor noise (the controller sees the truth).

	X [10^6 cells/mL]		m [%]		G [mM]		L [mM]		DO [%]	
Day	A	B	A	B	A	B	A	B	A	B
0	0.50	0.50	45.0	45.0	20.0	20.0	0.0	0.0	100	100
3	1.37	1.14	33.6	34.5	17.3	17.5	5.2	4.4	83	86
5	3.50	2.52	25.0	26.4	16.8	17.5	5.4	4.1	59	70
7	9.08	5.73	18.0	19.4	14.0	16.1	9.6	6.2	49	50
9	19.97	11.94	13.2	14.2	6.3	11.6	21.9	13.4	50	49
10	26.82	16.36	11.5	12.2	2.1	8.3	28.7	18.7	50	50

How to read the run:

- **Days 0–1.** Cells sit in the lag. Fresh medium under 5% CO_2 starts at pH 7.44; the pH loop raises CO_2 to about 8% and brings pH to 7.20 within a few hours.
- **Days 1–3.** Exponential growth starts, glucose falls and lactate rises, boosted by the early burst $\beta(t)$.
- **Days 3–5.** Perfusion ramps from 0 to 1.5 VVD. Fresh medium washes lactate out and brings glucose back; the dip and bump in both curves are the ramp.
- **Day 5.5 (A) / 6.4 (B).** Oxygen demand catches up with what the minimum rock rate supplies; DO reaches its 50% setpoint and the DO loop takes over, raising rock rate and, after day 8 for patient A, O_2.
- **Days 9–10.** Patient A runs short of glucose (2 mM by day 10) even at 1.5 VVD, and lactate approaches K_{iL}. The dose keeps growing while the memory share keeps falling: the trade-off the project is about.

6 The inner control loop (day 2)

6.1 The PI controller: `pid.py`

A positional PI controller with output limits and anti-windup by clamping. At each step k, with setpoint r, measurement y_k and interval Δt:

$$e_k = \sigma\,(r - y_k), \qquad \sigma = \begin{cases} +1 & \text{direct acting} \\ -1 & \text{reverse acting} \end{cases} \tag{20}$$

$$I_k = I_{k-1} + K_p\,\frac{\Delta t}{T_i}\,e_k, \tag{21}$$

$$\tilde{u}_k = u_{\text{bias}} + K_p\,e_k + I_k, \tag{22}$$

$$u_k = \min\big(\max(\tilde{u}_k,\ u_{\min}),\ u_{\max}\big), \tag{23}$$

$$I_k \leftarrow I_k - (\tilde{u}_k - u_k). \tag{24}$$

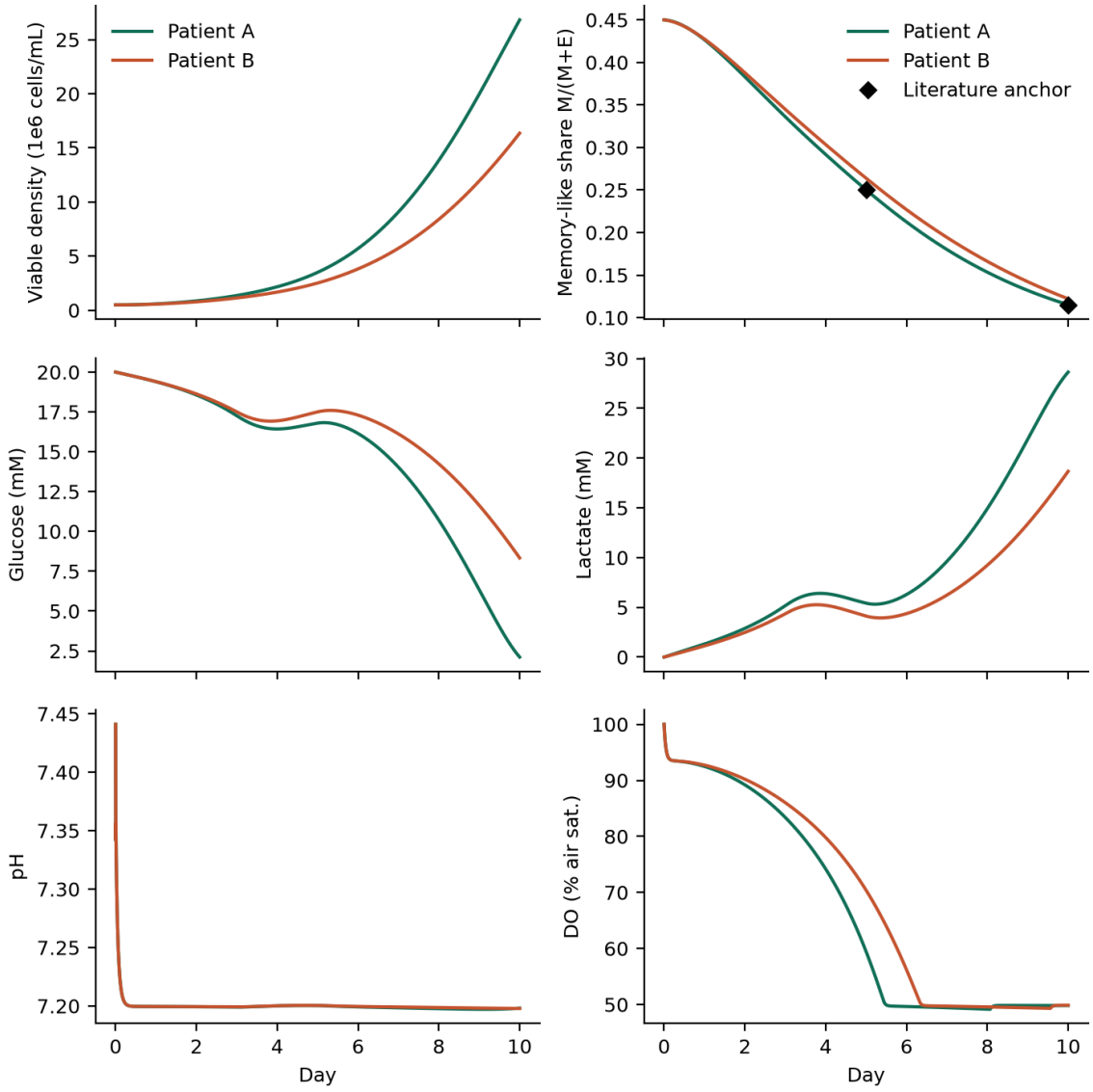


Figure 3: Patients A and B over ten days. Black diamonds: the literature memory anchors. pH and DO are under closed-loop control (section 6).

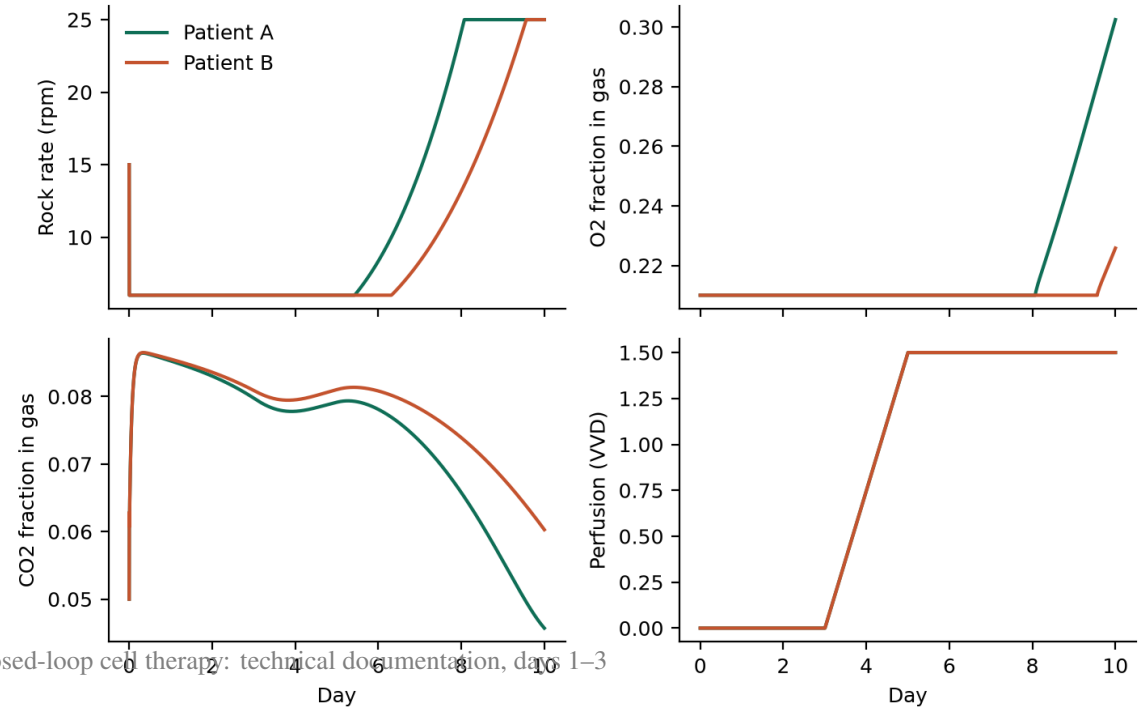


Figure 4: What the controller did: rock rate, O_2 and CO_2 fractions in the gas, and the perfusion schedule. Rock rate

The last line pulls the integral back to exactly where the output just reaches its limit, so a saturated loop recovers immediately once the error changes sign.

6.2 The rocker controller: `rocker.py`

Loop	Manipulated	Tuning	Notes
pH (setpoint 7.20)	$y_{CO_2} \in [0.01, 0.10]$	$K_p = 0.04$, $T_i = 0.25$ h, bias 0.05	reverse acting: more CO_2 lowers pH. No base addition is modelled.
DO (setpoint 50%)	$u_{DO} \in [0, 100]$	$K_p = 1.5$, $T_i = 1$ h, bias 20	split range onto rock rate and O_2, see (25).
Perfusion	$VVD \in [0, 2]$	schedule	see (26).

The DO output is split so that the rocker rocks harder first (gentler on T cells) and only then enriches the gas with oxygen:

$$(n,\ y_{O_2}) = \begin{cases} \left(n_{\min} + (n_{\max} - n_{\min})\,\dfrac{u_{DO}}{50},\ \ 0.21\right), & u_{DO} \le 50, \\ \left(n_{\max},\ \ 0.21 + (0.60 - 0.21)\,\dfrac{u_{DO} - 50}{50}\right), & u_{DO} > 50, \end{cases} \tag{25}$$

with $n_{\min} = 6$ rpm and $n_{\max} = 25$ rpm. Perfusion follows a schedule in culture days $t_d = t/24$:

$$VVD(t_d) = \begin{cases} 0, & t_d < 3, \\ 1.5\ \min\left(1,\ \dfrac{t_d - 3}{2}\right), & t_d \ge 3. \end{cases} \tag{26}$$

Hard limits (`Limits`): rock rate 6–25 rpm, O_2 21–60%, CO_2 1–10%, perfusion 0–2 VVD, pH setpoint 6.9–7.4, DO setpoint 20–80%. A supervisory layer (later, the digital twin and the operator) can call `set_setpoints()`, but the values are clipped to these limits inside the rocker, so no higher layer can push the bag outside them.

DO can only be raised, not lowered: there is no nitrogen. While the culture is young, the quasi-steady oxygen balance from (18g),

$$O_{ss} = C^* - \frac{\mathrm{OUR}}{k_L a(n_{\min})}, \tag{27}$$

lies above setpoint, so DO stays high with the rocker at its minimum and the DO loop saturated at zero.

7 The batch runner and the clock

7.1 `Batch` (`runner.py`)

`Batch` owns the state vector, the current inputs, the controller, optional sensors and the recorded history. One call to `step()`:

1. computes the true observables at t_k;
2. if sensors are attached, reads them; the controller then sees the readings $\hat{y}_k$, otherwise the truth y_k;
3. asks the controller for new inputs $u_k = C(\hat{y}_k)$;
4. integrates $\dot{x} = F(t, x, u_k)$ from t_k to $t_{k+1} = t_k + \Delta t$ with $\Delta t = 5$ min, using scipy's LSODA (rtol $= 10^{-6}$, atol $= 10^{-9}$), and clips the result at zero;
5. records a row with time, raw states, all observables, inputs, setpoints, sampling totals and the latest sensor readings.

`run(days)` repeats this; `run_batch(patient, days)` is the one-line version. `series(key)`, `at_day(day)` and `to_csv(path)` read the history back.

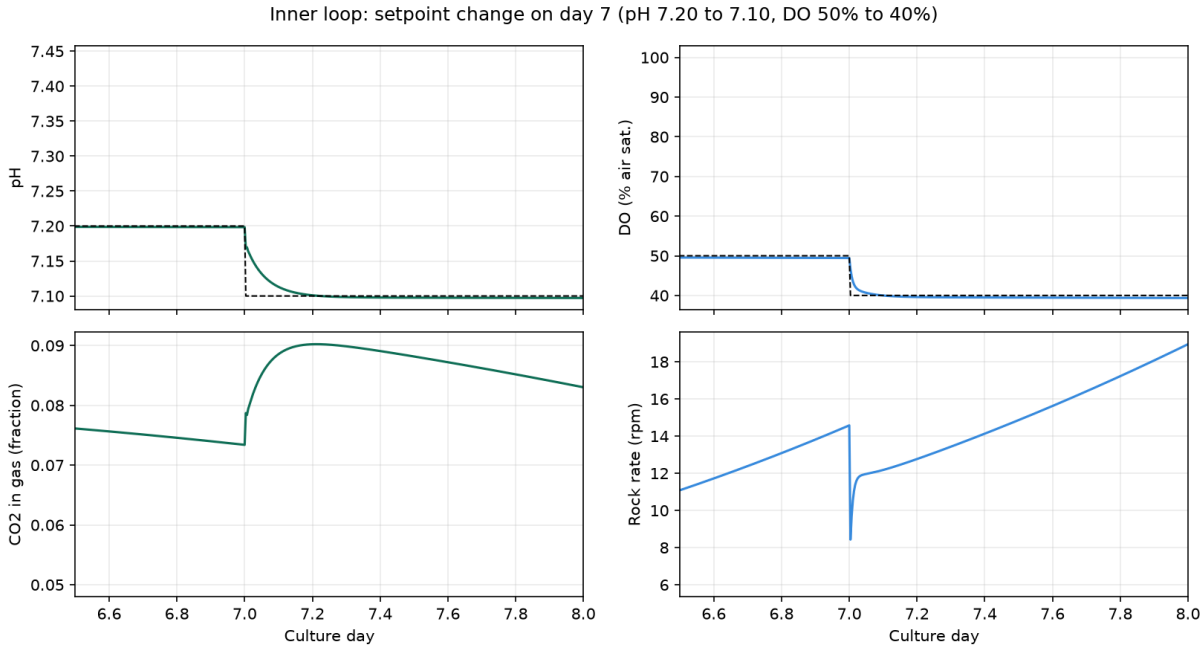


Figure 5: Step response (`scripts/step_response.py`): on day 7 the supervisory layer moves pH from 7.20 to 7.10 and DO from 50% to 40%. Both settle within a few hours without overshoot; the slow drift in CO_2 and rock rate afterwards follows the growing culture.

Sampling. `take_sample(volume_mL)` models a closed sample through the syringe port. It returns the true state of the drawn liquid; then the scale-controlled feed refills the bag with fresh medium. With $\phi = \min(V_s/V,\ 0.5)$:

$$(M, E, X_d) \leftarrow (1-\phi)\,(M, E, X_d), \qquad c \leftarrow c + \phi\,(c_{\text{in}} - c) \quad \text{for } c \in \{G, L, B\}, \tag{28}$$

and O is unchanged. The cells removed, $X\,V_s$, are added to `cells_sampled`, because every sample is taken from the patient's dose.

7.2 `SimClock` (`clock.py`)

Maps wall-clock time to simulated time at `speed_h_per_s` simulated hours per second (default 1: a nine-day run takes 3.6 minutes), with pause and resume. Nothing uses it yet; it is for the live devices and dashboard in later steps.

8 The sensor layer (day 3)

The model knows the true state; a real platform does not. The sensor layer reproduces how each quantity reaches the platform: how often, how noisy, how delayed and how it fails. All values are placeholders typical of single-use rocking-bag systems.

Signal	Source	Update	Noise	Drift	Delay
pH	optical patch	10 s	0.01 (res. 0.01)	−0.005/day	none
DO %	optical patch	10 s	1 (res. 0.1)	+0.5/day	none
bag weight	load cell	1 s	2 g	none	none
X, viability	at-line counter, 2 mL sample	on request	8% CV; 0.02	none	15 min
G, L	metabolite analyser	on request	3%, 4% CV	none	30 min
m	flow cytometry at harvest	once	0.02	none	12 h
OUR	soft sensor	each step	from k_La error	from DO	30 min window
acid prod.	soft sensor	each step	large	from pH	6 h window

8.1 Online sensors: `online.py`

For a channel with noise σ, drift rate ρ per day, resolution Δ and update period T_s, a new raw value at time t is

$$\hat{y}(t) = \Delta \cdot \mathrm{round}\left(\frac{y(t) + \rho\, t/24 + \varepsilon + \phi_f(t)}{\Delta}\right), \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \tag{29}$$

held until the next update $t + T_s$; each reading also reports its age. The fault term ϕ_f is zero unless a fault is injected from a start time t_f: **bias** ($\phi_f = c$), **drift** ($\phi_f = c\,(t - t_f)/24$), **stale** (value frozen) or **dropout** (no value, as after a lost connection). Faults will be used by the watchdog on day 9.

A drifting sensor moves the real process

The controller holds the *reading* at setpoint: $\hat{\mathrm{pH}} \approx 7.20$. With $\rho = -0.005$/day, the true pH therefore settles at $7.20 + 0.005\,t_d$, about +0.045 by day 9 (top left of figure 6; tested by `test_controller_follows_the_drifting_sensor`). The DO patch drifts the other way, so true DO ends a few percent below setpoint.

8.2 At-line analysers and flow cytometry: `atline.py`

`AtLineLab.submit()` takes the truth of a drawn sample and queues one result per instrument with

$$\hat{c} = \max\bigl(0,\ c\,(1+\varepsilon)\bigr),\ \ \varepsilon \sim \mathcal{N}(0, \mathrm{CV}^2) \quad \text{(counts, concentrations)}, \qquad \hat{v} = \mathrm{clip}_{[0,1]}(v + \varepsilon) \quad \text{(viability)}, \tag{30}$$

ready at $t_{\text{drawn}} + t_{\text{turnaround}}$. `poll(now)` releases the results that have become ready. `FlowCytometry.measure()` returns $\hat{m} = \mathrm{clip}_{[0,1]}(m + \varepsilon)$ with $\sigma = 0.02$, ready 12 h later: the delayed, expensive label the whole project is about.

8.3 Soft sensors: `soft.py`

Two rates the EnKF will use between at-line samples, computed from what the platform logs. Both use causal trailing windows of length w, as they would run live: $\overline{(\cdot)}_w$ is the moving mean and $\mathcal{S}_w(\cdot)$ the least-squares slope over $[t - w, t]$, reported only once the window holds at least 80% of w and three points.

8.3.1 Oxygen uptake rate ($w = 0.5$ h)

Rearranging (18g) for OUR and replacing true values with measured ones:

$$\widehat{\mathrm{OUR}}(t) = \overline{\left[\,\beta_k\, k_L a(n,\theta)\,(C^* - \hat{C})\,\right]}_w - \mathcal{S}_w(\hat{C}), \qquad \hat{C} = \frac{\widehat{\mathrm{DO}}}{100}\, C^*_{\mathrm{air}}, \tag{31}$$

where β_k (`kla_bias`) is a multiplicative error of the k_La correlation for this particular bag ($\beta_k = 1$ by default). OUR is proportional to cell number weighted by phenotype, so it is the best continuous proxy for growth.

8.3.2 *Acid production* ($w = 6$ h)

Inverting Henderson–Hasselbalch (6) gives bicarbonate from the pH reading and the gas:

$$\hat{B} = 10^{\widehat{\mathrm{pH}}-\mathrm{p}K_a}\; s\,\kappa\, y_{\mathrm{CO_2}}. \tag{32}$$

Rearranging (18f) for r_L:

$$\hat{r}_L(t) = -\frac{1}{\gamma}\Big[\, \mathcal{S}_w(\hat{B}) - \overline{D\,(B_{\mathrm{in}} - \hat{B})}_w \,\Big]. \tag{33}$$

The window is long because pH has a resolution of 0.01: one quantisation step in pH is a 2.3% step in $\hat{B}$, which swamps the slope over short windows.

9 Tests and calibration status

File	Test	What it checks
`test_process`	`states_stay_physical`	no negative states, viability ≤ 1
	`memory_share_passes_literature_anchors`	patient A within ± 0.03 of 25% (day 5) and 11.5% (day 10)
	`dose_rises_while_memory_falls`	dose up and m down from day 3 to 9
	`slow_donor_has_30_to_50_percent_fewer_cells`	X_B/X_A at day 9 in $[0.5, 0.7]$
	`inner_loop_holds_pH_and_DO`	after DO first reaches setpoint: $\lvert\mathrm{pH} - 7.2\rvert < 0.05$, $\lvert\mathrm{DO} - 50\rvert < 5$
`test_control`	`pi_holds_first_order_setpoint`	PI tracks a first-order plant
	`pi_anti_windup_recovers_quickly`	clamping prevents windup
	`perfusion_schedule_ramps`	schedule (26)
	`supervisory_setpoints_are_clipped`	`set_setpoints` respects hard limits
`test_sensors`	`online_noise_and_drift`	noise level and drift sign
	`reading_is_held_between_updates`	sample-and-hold at T_s
	`faults_dropout_and_stale`	dropout gives no value; stale freezes
	`atline_results_arrive_after_turnaround`	results released only after turnaround
	`sampling_costs_cells`	sampling dilutes the bag and counts cells removed
	`flow_cytometry_is_delayed`	flow result ready 12 h after drawing
	`controller_follows_the_drifting_sensor`	true pH rises as the patch drifts down
	`soft_oxygen_uptake_tracks_truth`	$\widehat{\mathrm{OUR}}$ close to the true OUR
`test_clock`	`clock_runs_pauses_and_jumps`	speed, pause and resume

Calibration (`docs/calibration.md`). The two memory anchors and the slow-donor gap are met. To digitise next: growth curves (Costariol 2019, Egan 2025) for $\mu_{\max}$ and τ_{lag}; specific glucose, lactate and oxygen rates (Costariol 2019, SMART 2026); the early-lactate correlation for a and b; the day-0 phenotype m_0 (Cadinanos-Garai 2025); the buffer share γ from Egan 2025 pH traces; and $k_L a_{\mathrm{ref}}$ for the bag. Digitised series go under `data/literature/<paper>/<figure>.csv`.

10 Observations, limitations and open points

Points noticed while writing this document. None breaks a test; they are worth knowing before the EnKF and calibration steps.

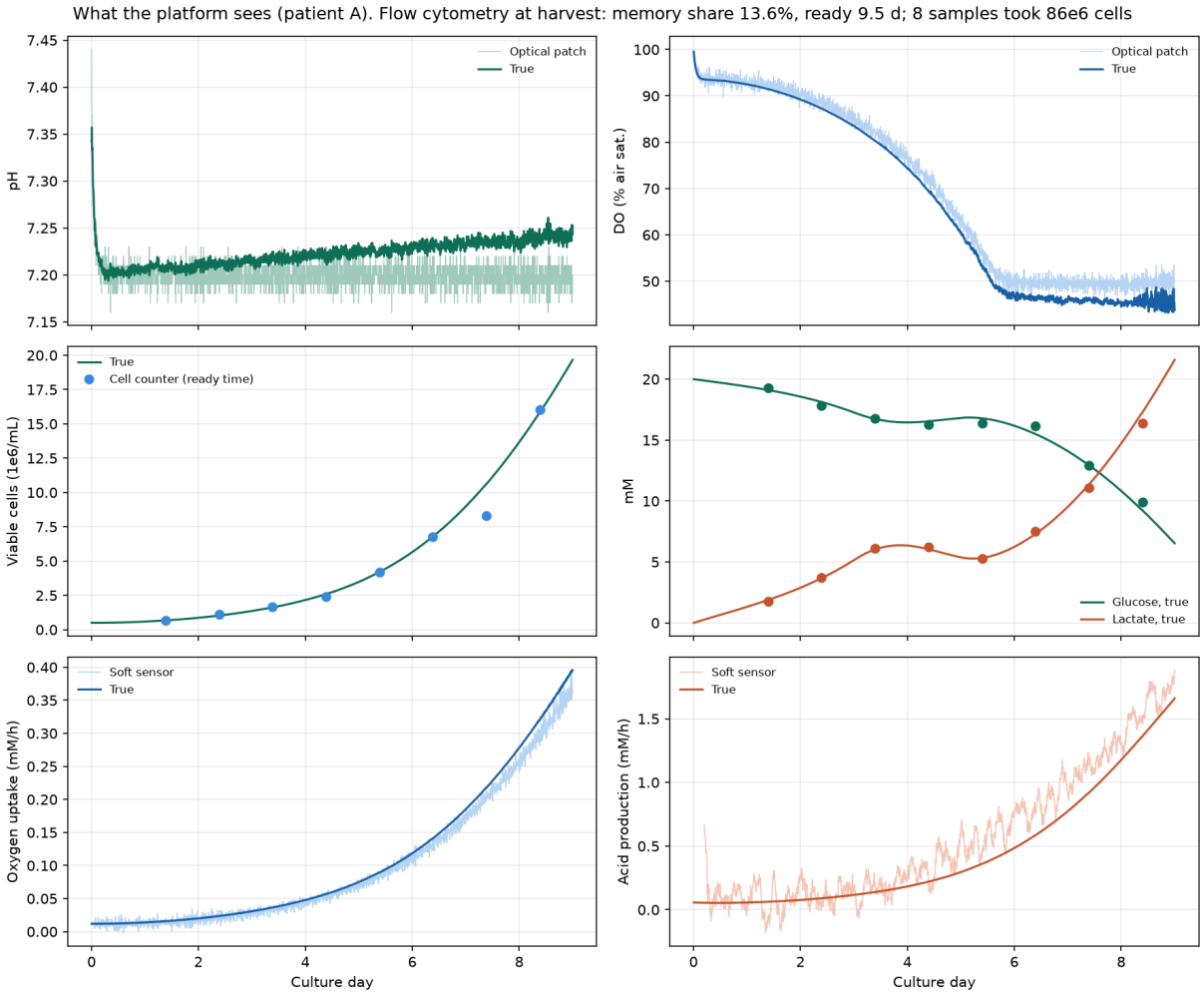


Figure 6: What the platform sees for patient A (`scripts/sensor_demo.py`): online patches against the truth, eight daily at-line samples plotted at their ready times, the two soft sensors, and flow cytometry at harvest. The eight 2 mL samples removed 86×10^6 cells.

1. **Patient B ends with a slightly higher memory share than A** (12.2% versus 11.5% at day 10), although B has the weaker activation, which by the hypothesis should mean less memory. Equation (19) explains it. B's conversion term is larger by

$$\frac{1 - 0.5 \times 0.25}{1 - 0.5 \times 0.5} = \frac{0.875}{0.75} \approx 1.17,$$

   but B's selection term is smaller, because $\mu_{\max,E} - \mu_{\max,M}$ is 0.0084 for B against 0.0101 for A and B's f is lower. Selection wins. If the intended story is "weaker activation, less memory", either r_a must be larger or the growth gap smaller. Either way, an estimator will have to separate growth speed from activation, which is exactly the hard part.

2. **DO is uncontrolled for the first 5–6 days.** With no N_2 and a 6 rpm floor, DO stays between 100% and 50%. Realistic for a young culture, but early DO and OUR are then governed by $k_L a(n_{\min})$, a placeholder.

3. $K_{O_2} = 0.005$ mM **is hard-coded** in `rates()` rather than in `ProcessParams`, unlike every other constant.

4. **pH acts only on growth**, not on metabolism or differentiation, and the pH loop can only add acid (CO_2), not base. If lactate overwhelms the buffer late in a dense run, the controller's only remedy is lowering CO_2 to 1%.

5. **Metabolism is not coupled to growth.** q_G and OUR depend on cell number and phenotype but not on f_{lag}, f_{pH} or f_L: cells in lag consume at full rate.

6. **Dead cells are retained and never lyse**, and the volume is constant apart from sampling, so viability only falls slowly and there is no late-run viability crash.

7. **The anchors are met by construction**: four parameters fitted to two points plus a density. They confirm the fit, not the model structure.

8. **`acid_production` is the lactate production rate** r_L, not the bicarbonate consumption γr_L. The soft sensor divides by γ to match; keep this in mind when comparing with literature acid rates.

11 What comes next

Day	Step	How it uses what exists
4	Rocker as an OPC UA server	exposes pH, DO, weight, rock rate, gas and pressure as industrial tags
5	Analyser as a REST API, data collector	wraps `AtLineLab`; a collector pulls both devices
6	EnKF soft sensor, sampling by information gain	(18) as the forecast model; OUR and acid production as continuous data, at-line results as sparse data; patient parameters augmented into the state
7	Run state machine and scheduler	orchestrates sampling and phases
8	Harvest recommendations, first dashboard pages	advisory decisions an operator approves; setpoints pass through the rocker's limits
9	Faults, watchdog, audit log	uses the fault injection in `online.py`; SQLite historian
10	Patient B page, Bayesian optimisation	campaign loop across process-development runs
11–12	Validation, packaging, rehearsal	